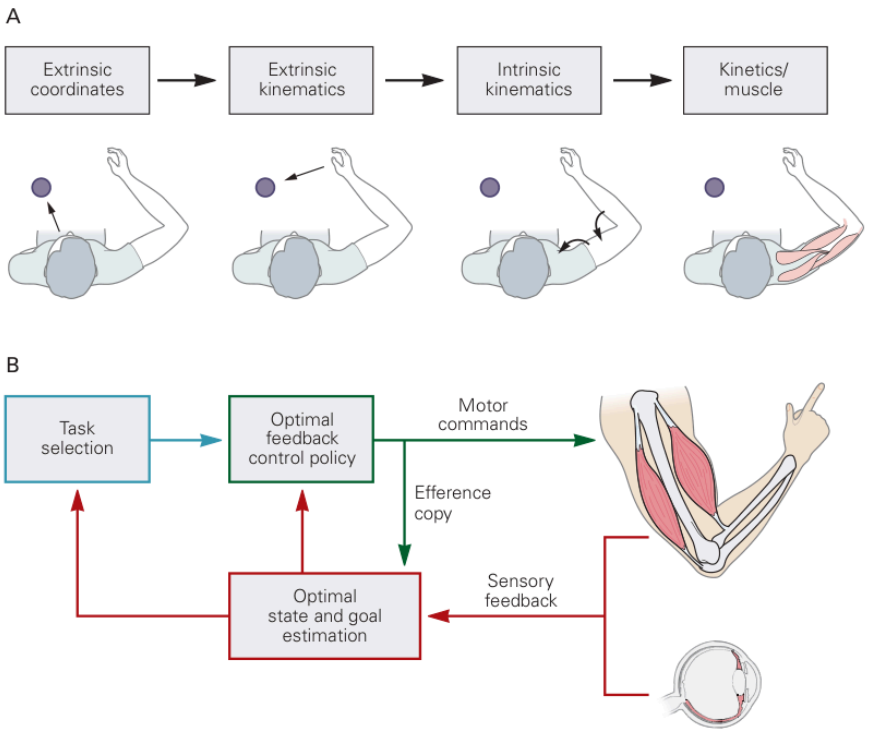
PHYSIOLOGY OF THE MOTOR SYSTEM

Writing in conversation with the chapter

Lecture 02 · writing workshop

The question



Kandel 6e, Fig. 34–1, p. 817

Which of the two frameworks in Figure 34–1 do you find more useful for understanding voluntary movement, and what important aspect of motor control does it leave unexplained?

150 words maximum

- A: a chain of coordinate transformations
- B: goal-dependent feedback control

What we wrote

Everyone chose B.

The gaps we named:

- How neurons implement it
- Where the cost function and internal model come from
- How competing objectives are weighed

Few answers used evidence from the chapter beyond Figure 34–1.

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Few answers used evidence from the chapter beyond Figure 34–1.

> Optimal feedback control "explains certain features of motor behavior but is largely agnostic as to the neural implementation for control." (p. 818)

> First check: does the chapter already make my point? If so, go one step further.

What Kandel says, and how the chapter is built

WHAT IT SAYS

> "This serial view … has long dominated neuroscience; this textbook, for example, has separate sections dedicated to perception, cognition, and movement." (p. 816)

> Framework A "places all emphasis on strictly serial feedforward computations." (p. 817)

> The two frameworks "are not mutually incompatible hypotheses." (p. 818)

HOW IT IS BUILT

Parietal cortex
… for State Estimation to Plan and Execute Motor Actions

↓

Premotor cortex
Supports Motor Selection and Planning

↓

Primary motor cortex
Plays an Important Role in Motor Execution

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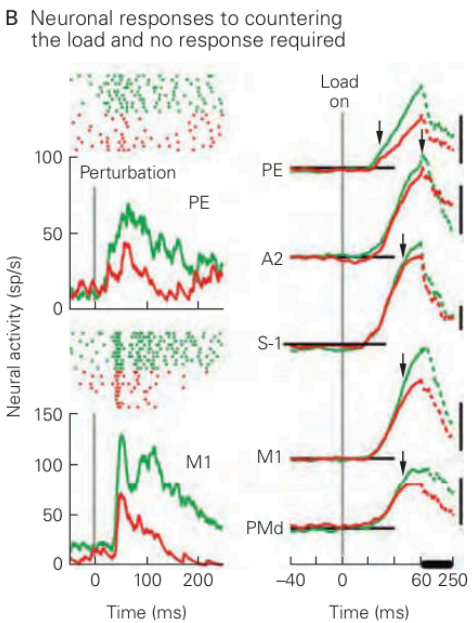
↓

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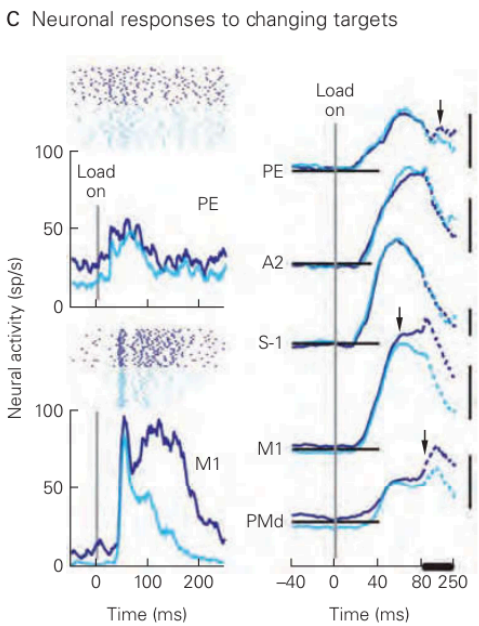
> The tension: B's processes are laid out along a serial anatomical route.

Evidence that can change the argument

Counter the load vs. watch a movie

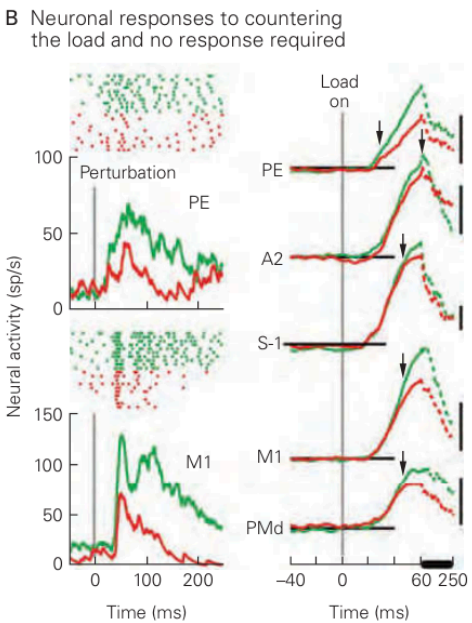


Load pushes toward vs. away from a new target

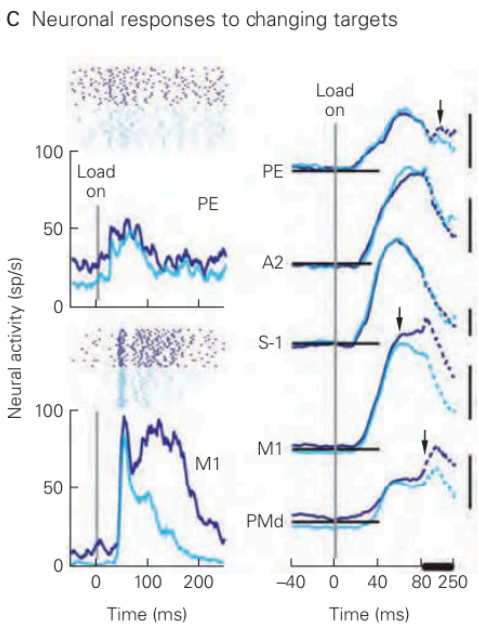


Evidence that can change the argument

Counter the load vs. watch a movie



Load pushes toward vs. away from a new target



All areas respond by about 20 ms in every condition.

The first area to differ is **PE** in one task and **M1** in the other.

PE: parietal area 5 · A2: somatosensory area 2 · S1: primary somatosensory cortex · PMd: dorsal premotor cortex · M1: primary motor cortex

Kandel 6e, Fig. 34–23B/C, p. 853 · arrows: first difference between conditions

Draft 1: 159 words, developed with ChatGPT

I find B more useful, but the organization of Chapter 34 invites us to read A into the anatomy: parietal cortex estimates the state of the world, premotor cortex selects an action, and M1 executes it.

Tension in the chapter

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I find B more useful, but the organization of Chapter 34 invites us to read A into the anatomy: parietal cortex estimates the state of the world, premotor cortex selects an action, and M1 executes it. However, while the structure of the chapter is serial, Figure 34–23 already challenges this progression by showing that the serial progression can be affected by details of the task: PE first when perturbations happen during a postural task and M1 first when the load cues a reach to another target.

Tension in the chapter **Evidence**

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Tension in the chapter Evidence Inference

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Tension in the chapter **Evidence** **Inference** **What B leaves unexplained**

Score the draft with the rubric

Score each criterion from 0 to 100 yourself, then compare with mine.

Criterion	Weight	Score	Lost	Why
English accuracy	2			
Style	1			
Scientific accuracy	2			
Insightfulness	3			
AI-relative originality	1			
Class-relative originality	1			

Rubric: Writing and Grading Policy 2027-1, version 1.1

Score the draft with the rubric

Score each criterion from 0 to 100 yourself, then compare with mine.

Criterion	Weight	Score	Lost	Why
English accuracy	2	**90**	2.0	Clear; “schema A” and “framework A” mixed
Style	1	**75**	2.5	Over 150 words; “serial” three times; “However, while”
Scientific accuracy	2	**70**	6.0	Figure read correctly, but B’s processes called A; “certainly” overstates
Insightfulness	3	**85**	4.5	Anchored feedback-or-prediction inference; gap close to p. 818
AI-relative originality	1	**70**	3.0	Uncommon start and inference; Fig. 34–23 is a common choice
Class-relative originality	1	**85**	1.5	No classmate used the chapter’s structure

> Weighted grade 80.5 / 100. Most points are lost on scientific accuracy.

Rubric: Writing and Grading Policy 2027-1, version 1.1

Ideas for improvement

Scientific accuracy · −6

The chapter lays B’s processes (state estimation, selection, execution) along an anatomical relay; A is a chain of coordinate frames. Name what is serial correctly and soften “certainly”: A says nothing about which area responds first.

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Style · −2.5

Cut to 150 words: use "serial" once, choose "However" or "while", and split the second sentence.

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Originality

Keep the tension in the chapter's structure. It is the idea no classmate wrote.

Your turn: revise your answer

1. Open your submitted answer.
2. Choose one Chapter 34 claim or figure that influences the fundamentals of your argument:
   - Supports
   - Challenges
   - Illuminates
   - Exposes a gap
3. AI is welcome.
4. Consider where best to incorporate that claim or figure into your argument.
5. Share with the group

> 10 minutes to consider.
>
> Then we will discuss.